Akoizo Original Practice - 2027 Rotation Form 1

Division B | Water Quality | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Explain why this transect is an estuary rather than a freshwater-only habitat.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
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2. (2 points) Explain how a nutrient pulse can lead to lower overnight oxygen.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
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3. (2 points) Identify one watershed action that could reduce nutrient loading and explain its pathway.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
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4. (2 points) Explain how traditional ecological knowledge could improve this monitoring plan without replacing measured evidence.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
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5. (2 points) Why should a reef food-web analysis distinguish herbivores from predators?

Answer: _____
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6. (2 points) Distinguish a harmful algal bloom from an ordinary observation of green water.

Answer: _____
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7. (2 points) A reef fish has fused beak-like teeth and scrapes algae from hard surfaces. Identify its common-name group.

Answer: _____
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8. (2 points) A starfish has many arms and venomous spines and consumes live coral tissue. Identify it.

Answer: _____
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9. (2 points) A reef-building animal secretes a hard calcium carbonate skeleton. Identify its listed group.

Answer: _____
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10. (2 points) A dark urchin has conspicuously long slender spines and grazes algae. Identify its listed group.

Answer: _____
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11. (2 points) A shrimp has red-and-white bands and conspicuous elongated appendages. Identify the listed common-name group.

Answer: _____
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12. (2 points) An elongate, soft-bodied echinoderm processes sediment on the seafloor. Identify its listed group.

Answer: _____
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13. (2 points) Calculate the overnight dissolved-oxygen decline.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
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14. (2 points) Calculate the BOD-bottle oxygen demand under the stated assumptions.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
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15. (2 points) A mixture contains equal volumes of freshwater and seawater with salinity 22. Assuming additive volumes and conservative mixing, estimate salinity.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

Answer: _____
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16. (2 points) Aragonite saturation decreases while other factors are held fixed. Explain a relevant concern for stony corals.

Answer: _____
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17. (2 points) Explain why high turbidity and high dissolved-solids concentration are not interchangeable observations.

Answer: _____
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18. (2 points) A field pH value changes between daytime and predawn. Name a biological mechanism consistent with this and one check on the inference.

Answer: _____
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19. (2 points) A salinometer calibration gives reading 10 at 2% mass/volume salt and reading 30 at 6%. A sample reads 16. Use linear interpolation to estimate concentration.

Answer: _____
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20. (2 points) How many grams of salt are required for 500 mL of final solution at 3% mass/volume?

Answer: _____
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